🌟 Graph Theory & Combinatorics Solutions 2022

🌟 Q1 (i) Show that the number of vertices of odd degrees in a graph is always even with a suitable example.

✨ **Answer:**

🔹 **Theorem:** In any undirected graph, the number of vertices having an odd degree is always even.

🧠 **Proof using Handshaking Lemma:** We know the famous Handshaking Lemma:

$$\sum \deg(v)=2|E|$$

Here:

- $\deg(v)$ = degree of a vertex
- $|E|$ = number of edges

Since the right-hand side is multiplied by **2**, the sum of all vertex degrees is always an even number! ✅

🔹 **Now Separate the Vertices:** Suppose:

- Some vertices have an even degree.
- Some vertices have an odd degree.

👉 Sum of all even degrees = Even number 👉 Total sum of all degrees = Even number

Therefore: 👉 The sum of odd degrees must also be even.

But mathematically:

- Sum of an even number of odd numbers = Even ✅
- Sum of an odd number of odd numbers = Odd ❌

Hence, the number of odd-degree vertices must absolutely be even.

✅ **Hence Proved.**

🌈 **Suitable Example:** Consider the following graph:

```
A ----- B
|       |
|       |
D ----- C
```

Degrees:

Vertex	Degree
A	2
B	2
C	2
D	2

Odd degree vertices = **0**. Since **0** is an even number, the theorem holds! ✅

🔥 **Another Example:**

```
A ----- B ----- C
```

Degrees:

Vertex	Degree
A	1
B	2
C	1

Odd degree vertices:

- A
- C

Total odd degree vertices = **2** ✅ (Which is an even number!)

🎯 **Final Conclusion:** ✨ Therefore, in every graph, the number of vertices having an odd degree is always even.

🌟 Q1 (ii) In a group of thirteen (13) persons is it possible that everyone has a friendship with exactly seven (7) in the group? Justify this.

✨ **Answer:**

Let:

- Number of persons = **13**
- Each person has a friendship with exactly **7** persons.

🧠 **Convert into Graph Theory:** Represent:

- Each person as a vertex 👤
- Friendship as an edge 🤝

Then:

- The degree of every vertex = **7**

So the total degree sum would be: $13 \times 7=91$

🔥 **Apply Handshaking Lemma:** According to the Handshaking Lemma:

$$\sum \deg(v)=2|E|$$

The sum of degrees of all vertices must be **EVEN** because it equals twice the number of edges. But here, **91** is **ODD** ❌. This is mathematically impossible.

❌ **Therefore:** A graph with:

- **13** vertices
- each having a degree of **7**

cannot exist. Hence, it is NOT possible that everyone has a friendship with exactly 7 persons.

✅ **Final Answer:** 🚫 No, it is not possible because the total degree sum becomes **91**, which is odd, but the sum of degrees in any graph must always be even.

🌟 Q2 (i) Find and draw the Hamiltonian Path and Hamiltonian Circuit of the graph given below (Figure 1) if possible. Justify this if not possible.

✨ **Answer:**

🔹 **Definition of Hamiltonian Path:** A Hamiltonian Path is a path that visits every vertex exactly once. 🛣️

🔹 **Definition of Hamiltonian Circuit:** A Hamiltonian Circuit is a Hamiltonian Path that starts and ends at the same vertex. 🔄

🔍 **From Figure 1:** The graph contains vertices arranged in a connected cyclic form with one extra pendant vertex attached on the right side.

⚠️ **Important Observation:**

- The rightmost vertex has a degree of **1** (it is a pendant vertex).

A Hamiltonian Circuit requires every vertex to have a degree of at least **2** because we must:

- enter the vertex
- leave the vertex

But the pendant vertex has only one edge.

❌ **Hamiltonian Circuit is NOT Possible** Because:

- The pendant vertex has a degree of **1**.
- The circuit cannot return back after visiting it.

Hence, the graph is not Hamiltonian.

✅ **Hamiltonian Path is Possible** One Hamiltonian Path is: Pendant → Lower Right → Upper Right → Top → Upper Left → Lower Left → Bottom → Center

This perfectly visits every vertex exactly once. ✅

🎯 **Final Conclusion:**

Type	Possible?
Hamiltonian Path	✅ Yes
Hamiltonian Circuit	❌ No

Reason: The presence of a pendant vertex of degree **1** prevents the formation of a Hamiltonian Circuit.

🌟 Q2 (ii) Show that a Hamiltonian Path is a Spanning Tree using a suitable example.

✨ **Answer:**

🔹 **Hamiltonian Path:** A Hamiltonian Path visits all vertices exactly once.

🔹 **Spanning Tree:** A spanning tree:

- contains all vertices
- is connected
- contains no cycles

🧠 **Why a Hamiltonian Path is a Spanning Tree:** Suppose a graph has n vertices. A Hamiltonian Path:

- visits all n vertices exactly once
- uses exactly $n-1$ edges
- contains no cycles

These are exactly the core properties of a spanning tree! Hence: ✅ Every Hamiltonian Path forms a Spanning Tree.

🌈 **Example:**

```
A ----- B ----- C ----- D
```

Properties of this path:

- All vertices included ✅
- Connected ✅
- No cycle ✅
- Edges = **3** = $n-1$ ✅

Therefore, this path perfectly functions as a spanning tree.

🎯 **Final Conclusion:** A Hamiltonian Path is a spanning tree because it spans all vertices, is connected, and has no cycles.

🌟 Q2 (iii) Using Prim's Algorithm find the Minimal Spanning Tree of the graph given below (Figure 2).

✨ **Answer:**

🔹 **Prim's Algorithm:** Prim's Algorithm constructs a Minimum Spanning Tree (MST) by:

- Starting from any vertex
- Selecting the minimum weight edge each time
- Avoiding cycles

🧠 **Given Edge Weights:** From Figure 2:

Edge	Weight
Left – Right	3
Left – Bottom Right	4
Left – Bottom Left	5
Left – Top	6
Bottom Left – Bottom Right	7
Top – Right	8
Right – Bottom Right	3

🔥 **Step-by-Step Prim's Algorithm:**

- **Step** 1️⃣ Choose smallest edge: **3** Connect: Left → Right

- **Step** 2️⃣ Next smallest edge: **3** Connect: Right → Bottom Right

- **Step** 3️⃣ Next minimum edge without forming a cycle: **4** Connect: Left → Bottom Right ⚠️ *Wait! This forms a cycle (Left → Right → Bottom Right → Left)* ❌ So we strictly reject it.

- **Step** 4️⃣ Next minimum: **5** Connect: Left → Bottom Left

- **Step** 5️⃣ Next minimum: **6** Connect: Left → Top

Now all vertices are fully connected! ✅

🌟 **Minimum Spanning Tree Edges:**

- Left — Right = **3**
- Right — Bottom Right = **3**
- Left — Bottom Left = **5**
- Left — Top = **6**

🔥 **Total Weight:** \$3+3+5+6=17\$

✅ **Final Answer:** The Minimum Spanning Tree contains edges of weights: **3, 3, 5, 6**. Total minimum cost: 🌟 **17**

🌟 Q3 (i) Find the Vertex Connectivity and Edge Connectivity of the graph given below (Figure 3).

✨ **Answer:**

🔹 **Vertex Connectivity \$K(G)\$:** Vertex connectivity is the minimum number of vertices whose removal disconnects the graph.

🔍 **Observation in Figure 3:**

- The middle vertex acts like a bridge between the left and right sections.

If this central vertex is removed:

- The graph splits into two totally disconnected parts.

Hence: \$\$K(G)=1\$\$

🔹 **Edge Connectivity \$\lambda(G)\$:** Edge connectivity is the minimum number of edges whose removal disconnects the graph.

- Removing one connecting edge alone is not sufficient.
- But removing two connecting edges disconnects the graph.

Hence: \$\$\lambda(G)=2\$\$

✅ **Final Answer:**

Quantity	Value
Vertex Connectivity	1
Edge Connectivity	2

🌟 Q3 (ii) Find the Union of the two graphs G and G′ given below (Figure 4).

✨ **Answer:**

🔹 **Definition of Union of Graphs:** Union of two graphs \$G \cup G'\$ contains:

- all vertices of both graphs
- all edges of both graphs

🔥 **Given Graphs:** Graph \$G\$ Contains:

- vertices: \$v_1, v_2, v_3, v_4\$

Graph G' Contains:

- vertices: v_5, v_6, v_7

✅ **Union Graph:** Vertices: $$V(G \cup G')={v_1, v_2, v_3, v_4, v_5, v_6, v_7}$$

Edges:

- all edges from G
- all edges from G'

🎯 **Final Conclusion:** The union graph contains every vertex and every edge from both graphs together.

🌟 Q3 (iii) Explain the Fundamental Cut Sets with a suitable example.

✨ **Answer:**

🔹 **Fundamental Cut Set:** A fundamental cut set is:

- a minimal set of edges
- whose removal disconnects the graph
- associated with a spanning tree

🧠 **Important Idea:** For every branch of a spanning tree:

- removing that branch divides the tree into two parts
- edges reconnecting those parts form a fundamental cut set

🌈 **Example:** Consider the graph:

```
A ----- B
|       |
|       |
D ----- C
```

Choose spanning tree:

```
A ----- B
|
|
D ----- C
```

If edge AB is removed:

- The graph becomes disconnected. The edges reconnecting the partitions form the fundamental cut set.

✅ **Final Conclusion:** A fundamental cut set is the minimum set of edges associated with a branch of a spanning tree whose removal disconnects the graph.

🌟 Q4 (i) “A graph is 2-colorable if it is bipartite and every cycle has even length.” Justify this with a suitable example.

✨ **Answer:**

🔹 **2-Colorable Graph:** A graph is called 2-colorable if:

- we can color all vertices using only 2 colors 🎨
- no two adjacent vertices have the same color.

🔹 **Bipartite Graph:** A graph is bipartite if:

- vertices can be divided into two disjoint sets
- edges occur only between the two sets.

🧠 **Important Theorem:** ✅ A graph is bipartite iff every cycle has an even length. Reason:

- Odd cycles create a color conflict ❌
- Even cycles allow alternate coloring ✅

🌈 **Suitable Example:** Consider a cycle graph:

```
A ----- B
|       |
|       |
D ----- C
```

This is a C_4 cycle. Length of cycle: **4**, which is EVEN ✅

🎨 **2-Coloring:** Use:

- Red 🔴
- Blue 🔵

Vertex	Color
A	🔴
B	🔵
C	🔴
D	🔵

Adjacent vertices have different colors ✅ Hence:

- The graph is bipartite.
- The graph is 2-colorable.

❌ **Counter Example:** Consider a triangle:

```
   A
  / \
 B---C
```

Cycle length: **3**, which is odd ❌ Try 2-coloring:

- Impossible because one adjacent pair will inevitably get the same color.

Hence:

- Not bipartite.
- Not 2-colorable.

🎯 **Final Conclusion:** A graph is 2-colorable exactly when it is bipartite, and all cycles are even.

🌟 Q4 (ii) Find the Chromatic Number of the graph given in Figure 5.

✨ **Answer:**

🔹 **Chromatic Number:** The chromatic number $\chi(G)$ is the minimum number of colors needed to color a graph so that adjacent vertices receive different colors.

🔍 **Observation of Figure 5:** The graph contains many intersecting edges and triangles. Since triangles are present:

- at least **3** colors are needed.

🎨 **Coloring Process:** Use colors:

- 🔴 Red
- 🔵 Blue
- 🟢 Green

Color vertices carefully such that adjacent vertices differ. Possible assignment:

Vertex Type	**Color**
Corner vertices	🔴
Middle vertices	🔵
Central vertex	🟢

No adjacent vertices share the same color ✅

❌ **Can we use only 2 colors?** No. Because the graph contains triangles, and odd cycles cannot be 2-colored.

Hence: $$\chi(G) \neq 2$$

✅ **Final Answer:** $$\chi(G)=3$$ The chromatic number of the graph is: 🌟 **3**

🌟 Q4 (iii) Find out how many vertices can be matched using maximum matching in the bipartite graph of Figure 6.

✨ **Answer:**

🔹 **Maximum Matching:** A matching is a set of edges such that:

- no two edges share a common vertex.

Maximum matching means:

- the largest possible matching.

🔍 **Observation from Figure 6:** Top vertices: a, b, c, d, e Bottom vertices: v, w, x, y, z

Possible matches:

- $a \leftrightarrow v$
- $b \leftrightarrow w$
- $c \leftrightarrow x$
- $d \leftrightarrow y$
- $e \leftrightarrow z$

✅ **Maximum Matching:** Number of matched edges: **5** Each edge matches 2 vertices. Hence matched vertices: $2 \times 5=10$

🎯 **Final Answer:** Maximum number of matched vertices: 🌟 **10 Vertices**

🌟 Q5 (i) There are 50 microcomputers in a computer center. Each microcomputer has 30 ports. How many different ports to a microcomputer in the center are there?

✨ **Answer:**

🔹 **Apply Product Rule:** Each:

- microcomputer = 50 choices
- port = 30 choices

Using Product Rule: $50 \times 30=1500$

✅ **Final Answer:** There are: 🌟 **1500 different ports** in the computer center.

🌟 Q5 (ii) Suppose there are eight runners in a race. Gold, silver and bronze medals are awarded. How many ways can medals be awarded if there are no ties?

✨ **Answer:**

🔹 **Important Concept:** Order matters because: Gold $\neq$ Silver $\neq$ Bronze Hence, use permutations.

🧠 **Calculation:** Gold medal: **8** choices Silver medal: **7** choices Bronze medal: **6** choices

Using Product Rule: $8 \times 7 \times 6=336$

✅ **Final Answer:** Number of ways to award medals: 🌟 **336 ways**

🌟 Q5 (iii) Suppose a department contains 10 men and 15 women. How many ways are there to form a committee with six members if it must have more women than men?

✨ **Answer:**

🔹 **Condition:** Committee size = 6 More women than men means possible cases:

Women	Men
4	2
5	1
6	0

🧠 **Case 1: 4 Women and 2 Men** Choose:

- 4 women from 15
- 2 men from 10

$$\binom{15}{4}\binom{10}{2} = 1365 \times 45 = 61425$$

🧠 **Case 2: 5 Women and 1 Man**

$$\binom{15}{5}\binom{10}{1} = 3003 \times 10 = 30030$$

🧠 **Case 3: 6 Women and 0 Men**

$$\binom{15}{6} = 5005$$

🔥 **Total Ways:** $61425+30030+5005=96460$

✅ **Final Answer:** 🌟 Total number of committees = **96460**

🌟 Q6 (i) Find the generating function of the sequence ⟨1, 4, 9, 16, 25, ...⟩ in closed form.

✨ **Answer:**

The sequence is: $$1^2, 2^2, 3^2, 4^2, \dots$$

General term: $$a_n=n^2$$

🔥 **Closed Form Generating Function:** The generating function for squares is:

$$\sum_{n=1}^{\infty} n^2x^n=\frac{x(x+1)}{(1-x)^3}$$

✅ **Final Answer:** Generating function: $$\frac{x(x+1)}{(1-x)^3}$$

🌟 Q6 (ii) Find the sequence generated by the generating function $\frac{4x}{1-x}$

✨ **Answer:**

We know: $$\frac{1}{1-x}=1+x+x^2+x^3+\cdots$$

Multiply by $4x$: $$\frac{4x}{1-x}=4x+4x^2+4x^3+\cdots$$

🔥 **Generated Sequence:** $\langle 0, 4, 4, 4, 4, \dots \rangle$

✅ **Final Answer:** The sequence generated is: 🌟 **$\langle 0, 4, 4, 4, 4, \dots \rangle$**